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SUBJECT CODE NO: - RNP-8023
FACULTY OF SCIENCE AND TECHNOLOGY
S.E (EEE / EE / EEP) (NEP) Sem - III
Examination November / December 2025 (To be held in February-2026)
Electronics Devices & Circuits

[Time: 2 Hours]**[Max. Marks: 30]**

Please check whether you have got the right question paper.

N. B

- 1) Part A: Question No 1 is compulsory.
- 2) Attempt any FOUR questions from Part B: Q.2 to Q.7.
- 3) Use of non-programmable calculator is allowed

Part A**Q1 Attempt ANY FIVE of the following questions****10**

- a) Draw the Symbols BJT, FET & MOSFET
- b) Define Load Regulation & Line Regulation
- c) Draw the Drain & Transfer Characteristics of N-Channel FET"
- d) State the Barkhausen Criteria for Oscillations
- e) Match the pairs:

Sr. No.	Parameters	Sr. No.	Options
1	Crystal Oscillator	a	Removes Ripple
2	Voltage Divider Circuit	b	Produces High frequency signal
3	Voltage Multiplier	c	Universal Biasing
4	Filter circuit	d	Converts low AC to high DC voltage

- f) State the advantages & Disadvantages of negative feedback
- g) Define oscillator with Block diagram

Part B

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| Q. 2 Draw & Explain the Block diagram of DC regulated Power Supply, with wave-forms | 5 |
| Q. 3 Define a Filter and explain all types of filters with input & output waveforms. | 5 |
| Q. 4 Draw & Explain the circuit diagram of BJT based RC coupled amplifier | 5 |
| Q. 5 What is BJT Biasing? State the types of Biasing & explain any one type | 5 |
| Q. 6 Define an FET. Explain working of N-Channel FET with suitable diagrams | 5 |
| Q. 7 Define an oscillator? Explain Colpitt's Oscillator in details | 5 |